

Sampling biomass estimates from allometric equations

A1 Setup

We make use of these packages

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

`filter`, `lag`

The following objects are masked from 'package:base':

`intersect`, `setdiff`, `setequal`, `union`

```
library(ggplot2)  
library(cowplot)
```

A2 Allometric relationships

Gruner (2003) published coefficients for taxon- and stage-specific allometric relationships of the form

$$\text{mass} = a(\text{length})^b$$

This allometric relationship was estimated from real data based on the log-log transform and assuming a normal distribution of log-transformed mass

$$\begin{aligned}\widehat{\log(\text{mass})} &= \log(a) + b \log(\text{length}) \\ \log(\text{mass}) &\sim \mathcal{N}(\widehat{\log(\text{mass})}, \sigma)\end{aligned}$$

Gruner (2003) also published R^2 coefficients for these log-log transformed allometric relationships.

Gruner (2007) measured length of individual specimens to 1 mm resolution and used the estimated coefficients from these allometric relationships to approximate the mass of each specimen as $e^{\log(a)+b\log(\text{length})}$. This is a reasonable assumption but leads to potential issues when comparing data to Maximum Entropy Theory of Ecology [METE; Harte (2011)] predictions because METE assumes biomass-derived metabolic rates are continuous while using expected values based on 1 mm resolution length measurements introduces artificial discretization of the data.

A3 Drawing random biomass samples

To overcome this potential issue we conduct our METE analyses with random draws from $\log(\text{mass}) \sim \mathcal{N}(\widehat{\log(\text{mass})}, \sigma)$, one draw per individual. Random draws will be centered around $\widehat{\log(\text{mass})}$ but be continuous-valued rather than artificially discretized.

In order to achieve those random draws we must match the allometric coefficients from Gruner (2003) to the taxonomy and stage information in Gruner (2007). A modified version of Table 1 from Gruner (2003) is shown in Appendix Table 1

```
allom <- read.csv("../gruner2003.csv")
knitr::kable(allom)
```

Appendix Table 1: Coefficient estimates for the allometric relationship $\text{mass} = a(\text{length})^b$, modified from Table 1 of Gruner (2003)

taxon	stage	n	a	b	R2
Arthropoda	all	488	0.0340	2.191	0.773
Insecta	all	462	0.0350	2.173	0.762
Crustacea	all	26	0.0138	2.669	0.918
Isopoda	all	16	0.0152	2.770	0.974
Amphipoda	all	10	0.0088	2.682	0.989

Appendix Table 1: Coefficient estimates for the allometric relationship $\text{mass} = a(\text{length})^b$,
modified from Table 1 of Gruner (2003)

taxon	stage	n	a	b	R ²
Arachnida	all	46	0.0562	2.332	0.895
Araneae	all	30	0.0332	2.924	0.929
Tetragnathidae	all	13	0.0372	2.406	0.929
Blattodea	all	18	0.0313	2.358	0.955
Hymenoptera	adult	22	0.0139	2.383	0.918
Orthoptera	all	20	0.0180	2.720	0.919
Psocoptera	all	18	0.0136	3.115	0.786
Coleoptera	all	130	0.0336	2.347	0.771
Coleoptera	adult	123	0.0339	2.384	0.801
Carabidae	adult	16	0.0168	2.752	0.969
Scolytidae	adult	13	0.0313	2.531	0.736
Ciidae	adult	13	0.0313	2.531	0.736
Coccinellidae	adult	12	0.0724	2.665	0.651
Nitidulidae	adult	22	0.0679	1.308	0.384
Elateridae	adult	17	0.0138	2.595	0.959
Staphylinidae	adult	17	0.0138	2.595	0.959
Silvanidae	adult	17	0.0138	2.595	0.959
Curculionidae	adult	43	0.0310	2.641	0.887
Anobiidae	adult	43	0.0310	2.641	0.887
Aglycyderidae	adult	43	0.0310	2.641	0.887
Proterhinus	adult	19	0.0183	3.144	0.974
Heteroptera	all	55	0.0411	1.934	0.823
Lygaeidae	all	17	0.0375	2.065	0.861
Miridae	all	20	0.0320	2.318	0.762
Nabis	all	18	0.0071	2.764	0.935
Nesiomiris	all	18	0.0071	2.764	0.935
Homoptera	all	92	0.0234	2.536	0.846
Homoptera	adult	64	0.0155	2.835	0.905
Homoptera	immature	28	0.0585	1.670	0.627
Nesophrosyne	all	16	0.0067	3.141	0.884
Psyllidae	all	19	0.0175	2.629	0.839
Aphididae	all	19	0.0175	2.629	0.839
Psyllidae	adult	14	0.0123	2.995	0.752
Fulgoroidea	all	73	0.0323	2.338	0.787
Siphanta acuta	all	18	0.0088	3.182	0.875
Cixiidae	all	38	0.0511	2.072	0.809
Delphacidae	all	38	0.0511	2.072	0.809
Lepidoptera	adult	9	0.0179	2.318	0.922

Appendix Table 1: Coefficient estimates for the allometric relationship $\text{mass} = a(\text{length})^b$, modified from Table 1 of Gruner (2003)

taxon	stage	n	a	b	R2
Lepidoptera	immature	14	0.0271	1.769	0.683
Diptera	adult	19	0.0153	2.573	0.850
Collembola	all	9	0.0056	2.809	0.924
Neuroptera	adult	8	0.0070	2.739	0.911
Coleoptera	immature	23	0.0290	1.730	0.848
Diptera	immature	23	0.0290	1.730	0.848
Hymenoptera	immature	23	0.0290	1.730	0.848
Neuroptera	immature	23	0.0290	1.730	0.848

Now we match rows in `allom` to rows in the data from Gruner (2007) (which we call `arth`) based on taxonomy and stage. Matching follows a priority hierarchy from genus to class and from specific stage to all stages lumped.

```
arth <- read.csv("../gruner2007.csv")

#' function to match based on taxonomy and stage, prioritizing finest
#' resolution match possible. to be applied over rows of `arth` with `mapply`.
#' a few key assumptions are made:
#'   - `arth$stage` has no missing data (it does not)
#'   - `allom[, c("taxon", "stage")]` has no duplicates (it does not)
#'
#' @param genus genus column from `arth`
#' @param family family column from `arth`
#' @param order order column from `arth`
#' @param class class column from `arth`
#' @param stage stage column from `arth`
#' @param d data frame of allometric coefficients
#'
#' @returns integer ID of row in allometry data frame

match_allom <- function(genus, family, order, class, stage, d) {
  tax_levels <- c("genus", "family", "order", "class")

  # vector holding specific taxon name at each level
  taxvals <- c(genus, family, order, class) |>
    setNames(tax_levels)
```

```

tmatch <- taxvals %in% d$taxon

# if nothing matches, return arthropod all
if(sum(tmatch) == 0) {
  d_idx <- d$taxon == "Arthropoda" & d$stage == "all"
} else {

  t_idx <- which(tmatch) |> min()
  tax <- taxvals[t_idx]

  d_idx <- d$taxon == tax

  # multiple rows match so match stage
  if(sum(d_idx) > 1) {
    d_idx <- d_idx & (stage == d$stage)
  }
}

return(which(d_idx))
}

# apply to every row of `arth`
allom_idx <- mapply(match_allom,
                    arth$genus, arth$family, arth$order,
                    arth$class, arth$stage,
                    MoreArgs = list(d = allom))

# add allom info to `arth` data
arth$taxon_matched <- allom$taxon[allom_idx]
arth$stage_matched <- allom$stage[allom_idx]
arth$a <- allom$a[allom_idx]
arth$b <- allom$b[allom_idx]
arth$R2 <- allom$R2[allom_idx]

# add prediction to `arth`
arth$biom_pred <- arth$a * arth$lengthmm^arth$b

# where prediction is NA (due to missing length) fill in with estimate from
# Gruner 2007
arth$biom_pred[is.na(arth$biom_pred)] <- arth$ind_biom[is.na(arth$biom_pred)]

```

Now we are ready to draw biomass samples.

A4 Draw $\log(\text{biomass})$ using considering R^2

To draw a sample of biomass we have to know σ , i.e. the standard deviation of the residuals of the log-log allometry model. We can back-calculate σ from R^2 using sums of squares:

$$R^2 = \frac{SS_{\text{regression}}}{SS_{\text{regression}} + SS_{\text{residual}}}$$
$$\Rightarrow SS_{\text{residual}} = SS_{\text{regression}} \cdot \frac{1 - R^2}{R^2}$$

The residual standard deviation is connected to $SS_{\text{residuals}}$ by $\sigma = \sqrt{SS_{\text{residual}}/(n-2)}$ where n is number of specimens. To calculate $SS_{\text{regression}}$ and n we have to group the `arth` data according to which taxon and stage each row matched to

```
# record predicted log biomass for further calculations
log_biom_pred <- log(arth$biom_pred)

# vector to hold residual standard deviations
sigma_resid <- numeric(length(log_biom_pred))

# group by matching index
groups <- split(1:nrow(arth), allom_idx)

# loop over groups calculating n and sigma
for(g in names(groups)) {
  # rows in `arth` to consider
  idx <- groups[[g]]

  # predictions for these groups
  preds <- log_biom_pred[idx]

  # number specimens for this group
  n_g <- length(idx)

  if(n_g < 3) {
    # too few points to estimate variance, assume sigma = 0
    s <- 0
  } else {
    # sum of squares regression
```

```

SS_reg <- sum((preds - mean(preds))^2)

if(SS_reg == 0) {
  # occurs when all length values are equal or slope = 0,
  # assume sigma = 0
  s <- 0
} else {
  # calculate sigma of residuals
  R2_g <- allom$R2[as.integer(g)]
  SS_res <- SS_reg * (1 - R2_g) / R2_g

  s <- sqrt(SS_res / (n_g - 2))
}

# update sigma vector
sigma_resid[idx] <- s
}

# add sigma of residuals to `arth`
arth$sigma_resid <- sigma_resid

```

Finally a random draw of biomass can be added to `arth` by sampling from

$$\log(y) \sim \mathcal{N}(\log(\text{predicted biomass}), \sigma)$$

```

# set seed for consistent results
set.seed(123)

arth$biom_sim <- rnorm(nrow(arth), arth$biom_pred, arth$sigma_resid)

```

A5 Sanity check against existing `ind_biom`

The published biomass estimates from Gruner (2007) are retained in `arth$ind_biom`. If our random samples of biomass were successful then $(\text{biom_sim} - \text{ind_biom}) / \text{sigma_resid}$ should follow a normal distribution with mean = 0 and standard deviation = 1.

```

# create a new data frame with a column that should be N(0, 1)
normalized <- filter(arth, sigma_resid > 0) |>
  mutate(norm_01 = (biom_sim - ind_biom) / sigma_resid)

```

A few specimens ($n = 59$ out of 17024 total specimens) are outliers. The most reliable treatment for those outliers is to set them equal to their originally published biomass from Gruner (2007)

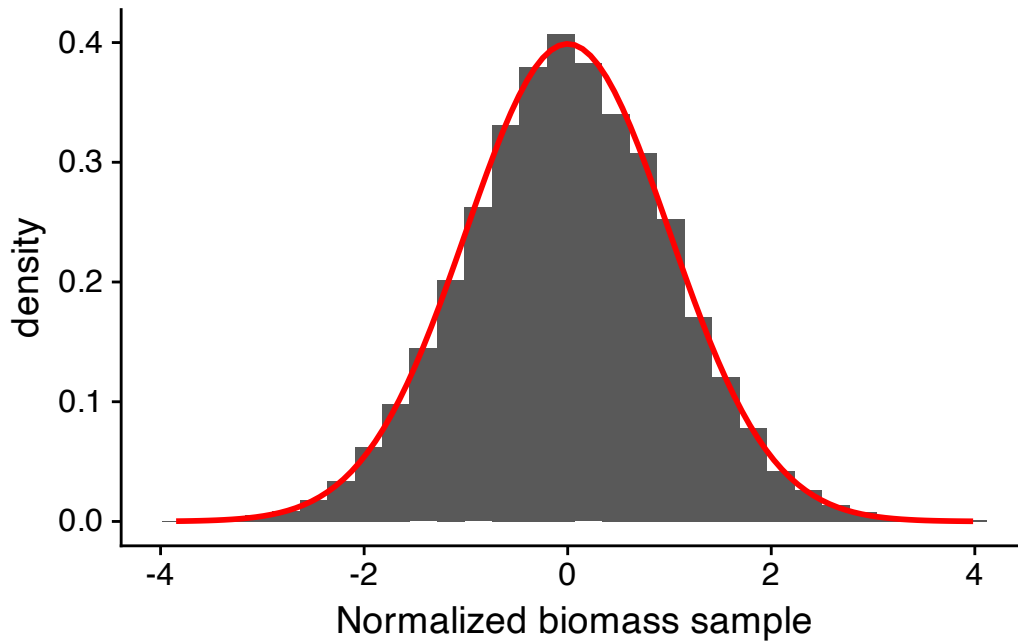
```
# find problem specimens
trouble <- filter(normalized, norm_01 > 4) |>
  select(specimen_code) |>
  pull() |>
  (\(x) arth$specimen_code %in% x)()

# set biomasses to original published
arth$biom_pred[trouble] <- arth$ind_biom[trouble]
arth$biom_sim[trouble] <- arth$ind_biom[trouble]

# ensure any future sampling will respect the original published value
arth$sigma_resid[trouble] <- 0
```

Now we can re-calculate the normalized samples and compare to $\mathcal{N}(0, 1)$ in Appendix Figure A1.

```
filter(arth, sigma_resid > 0) |>
  mutate(norm_01 = (biom_sim - ind_biom) / sigma_resid) |>
  ggplot(aes(norm_01)) +
  geom_histogram(aes(y = after_stat(density)), bins = 30) +
  stat_function(fun = dnorm, color = "red", linewidth = 1) +
  xlab("Normalized biomass sample") +
  theme_cowplot()
```

Appendix Figure A1: Checking normalized sample of biomass against expectation of $\mathcal{N}(0, 1)$ shown in red

The random draws indeed conform to the expectation.

A6 Saving the final product

The final computation is to add relative metabolic rate which we assume is simply

$$\text{metabolic rate} \propto \text{biomass}^{3/4}$$

METE normalizes metabolic rate such that the minimum is always 1, thus we only need to raise biomass to the $3/4$ power

```
arth$metabolic_rate <- arth$biom_sim^0.75
```

We ship these data with the R package *HawaiiArthMETE* so our last step is to save them to the `/data` directory.

```
save(arth, file = "../data/arth.rda")
```

References

- Gruner, D.S. (2003). Regressions of length and width to predict arthropod biomass in the hawaiian islands. *Pacific Science*, 57, 325–336.
- Gruner, D.S. (2007). Geological age, ecosystem development, and local resource constraints on arthropod community structure in the hawaiian islands. *Biological Journal of the Linnean Society*, 90, 551–570.
- Harte, J. (2011). *Maximum entropy and ecology: A theory of abundance, distribution, and energetics*. OUP Oxford.